# ICC CTCF Ölçer 2 — User Guide

*Version 2.0.0 · Erdal Balcan, Manisa Celal Bayar University · MIT license · DOI: 10.5281/zenodo.23860021*

Source code and downloads: https://github.com/erdal-b-dot/icc-ctcf-olcer · Türkçe kılavuz: `KULLANIM_KILAVUZU.md`

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## 1. What does the program do?

It measures **per-cell CTCF** (corrected total cell fluorescence) in immunocytochemistry (ICC) fluorescence images:

> **CTCF = Integrated density – (Cell area × Mean background)** (McCloy et al., 2014, *Cell Cycle* 13:1400–1412)

It automates the manual ImageJ workflow:

| **Manually in ImageJ** | **In this program** |
|---|---|
| Calibrate with Set Scale | The scale bar **and the value printed under it** (20 µm, 50 µm) are read automatically |
| Trace every cell with the polygon tool | Cells are segmented from FITC; touching cells are separated using DAPI nuclei |
| Draw and measure 3 background regions | 3 boxes are placed automatically away from cells (movable, or add your own) |
| Measure → copy to Excel → formula | Table, Excel, group summary, plots and statistics in one click |

**Measurements are identical to ImageJ/Fiji.** For the same cell boundaries, Area, Mean, StdDev, Min, Max, IntDen and RawIntDen are the same as Fiji's (largest difference < 0.00001 %). You can check this yourself from within the program: **Fiji ▸ Verify with Fiji** (Section 11).

Your images are **never sent anywhere**; the program works offline.

**Language:** the interface follows your computer's language (Turkish or English). Change it under **View ▸ Language / Dil** (takes effect after restart). Excel output is always in English.

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## 2. Installation

### 2.1 Ready-to-run app (recommended — no Python needed)

Download from **https://github.com/erdal-b-dot/icc-ctcf-olcer/releases** (newest version at the top).

| Computer | File | Minimum system |
|---|---|---|
| Apple Silicon Mac (M1/M2/M3/M4) | `ICC_CTCF_Olcer_<version>_mac_AppleSilicon.dmg` | macOS 12.3 |
| Intel Mac | `ICC_CTCF_Olcer_<version>_mac_Intel.dmg` | macOS 12.0 (Monterey) |
| Windows 10/11 (64-bit) | `ICC_CTCF_Olcer_<version>_Windows.zip` | Windows 10 |

> To find your Mac's processor: Apple menu ▸ **About This Mac**. "Chip: Apple M…" means Apple Silicon; "Processor: Intel…" means Intel.

**macOS:** 1. Double-click the `.dmg` → drag **ICC CTCF Ölçer** onto the **Applications** folder shown next to it. Then eject the disk (⏏) in the Finder sidebar. 2. **On first launch** do not double-click: **right-click the app ▸ Open ▸ Open**. (The app is not notarized by Apple, so macOS shows an "unidentified developer" warning; this is needed only once.) 3. If you see *"is damaged and can't be opened"* (happens with downloaded files), run once in Terminal: `xattr -cr "/Applications/ICC CTCF Ölçer.app"`

**Windows:** 1. Right-click the zip ▸ **Extract All**. Put the folder e.g. in `Documents` (do not run it from inside the zip). 2. Double-click `ICC CTCF Olcer.exe`. If *"Windows protected your PC"* appears: **More info ▸ Run anyway**. 3. Desktop shortcut: right-click the exe ▸ Send to ▸ Desktop (create shortcut).

### 2.2 Running from source

With a Python environment (to receive updates via `git pull`): - macOS: `./install_mac.sh`, then `./run_desktop.command` - Windows: `install_windows.bat`, then `start_desktop.bat`

Building the stand-alone packages yourself: `./build_mac.sh` (macOS) / `build_windows.bat` (Windows).

### 2.3 Fiji (optional but recommended)

The Fiji bridge (Section 11) requires Fiji: https://imagej.net/software/fiji/downloads The program looks for Fiji in `Downloads/Fiji`, `Applications/Fiji`, your home folder (`~/Fiji`, `C:\Fiji`). Elsewhere: **Fiji ▸ Choose Fiji location…** (asked once, remembered).

### 2.4 Earlier browser version (v1)

v1 (the browser-based Streamlit version, Turkish only) is still available under the `v1.0.0` tag. v1 and v2 give the same numbers for the same cell boundaries, but the **nucleus detection method differs**, so boundaries between touching cells are drawn slightly differently (Section 15). Measure all images of a study with **the same version**.

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## 3. Image requirements

However good the program is, the validity of the measurement depends on the image.

**Required:** - **Raw, single-channel images.** The FITC image saved directly by the microscope software (cellSens etc.). Images that went through Photoshop, had brightness/contrast adjusted, or were **merged cannot be used** for CTCF — the program detects these and warns you. - **All compared images acquired with identical settings:** same objective, exposure time, gain, lamp power, white balance. - **Same objective.** Do not compare 20× with 40× images — the program warns if objectives are mixed. - **No saturated pixels.** The number of saturated pixels is reported for every cell ("Saturated px" column).

**Recommended:** - **The DAPI image of the same field.** Essential for separating touching cells correctly (Warnings). - **Keep the scale bar in the image.** Calibration is read from it (cellSens style: white box in the lower right corner). - **TIFF** is better than JPEG (JPEG is lossy). Use the same format for all images of a study.

**Tested equipment:** Olympus BX43 fluorescence microscope, Olympus DP74 colour camera, cellSens Entry software. On this camera FITC appears yellow-green and DAPI **cyan**. FITC is measured on the **green (G) channel**; FITC and DAPI are told apart by the blue/red ratio of the brightest pixels. Other cameras/software should work for measurement; if the scale-bar label cannot be read, the program warns you and asks for the value (Section 6.2).

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## 4. Quick start (5 minutes)

1. Open the program.
2. In Finder/Explorer, **select the FITC and DAPI files together** and drag them onto the window. The program tells which one is DAPI by its colour.
3. Check the right panel: **Calibration** (e.g. 0.1449 µm/pixel, 20 µm, *read from label*), **Objective**, **Cell count**, the green **"Cell separation: DAPI nuclei"** box.
4. **Select all cells** (or click the cells to measure — they turn yellow). The **Cells** tab below shows CTCF.
5. In the right panel enter **Group / condition** (e.g. *Control*) and **Experiment / replicate** (e.g. *1*) → **+ Add to results table** (⌘↩ / Ctrl+Enter).
6. Next image: close with **⌘W**, drag the next pair, change the group, add.
7. When done: **Save to Excel** (⌘E) and **Plot…** (⌘G).

With many images, use **Batch-measure folder** (⌘B) instead of steps 4–6 (Section 8).

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## 5. The main window

| Part | Contents |
|---|---|
| **Menus** | File · Results · Fiji · Edit · View · Help |
| **Toolbar 1** | Open FITC · Open DAPI · Close images · Fit/Zoom in/Zoom out · **Display:** FITC / DAPI / Merged · Nucleus outlines |
| **Toolbar 2** | **Boundary editing:** Select · Split · Merge · Delete · Draw · Undo · Redo — **Results:** add to table · batch · Excel · Plot · Verify with Fiji |
| **Image area** | Mouse wheel: zoom · Drag: pan · Click: select cell |
| **Right panel** | Files, scale bar, calibration, objective, counts, cell-separation indicator, cell selection, background settings, results (group, experiment), legend, display brightness |
| **Bottom tabs** | Cells · Background boxes · Collected results · Group summary |

**Outlines on the image:**

| Outline | Meaning |
|---|---|
| Grey, numbered | Cell boundary found by the program |
| **Yellow** | Selected (measured) cell |
| **Magenta** | Nucleus boundary found in DAPI |
| **Red square** (BG1, BG2…) | Background box |

The **Display brightness** slider and **Display: FITC/DAPI/Merged** change **only the screen**; measurement always uses the raw FITC (green channel) data.

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## 6. Measuring a single image — step by step

### 6.1 Opening images

- **Drag and drop** (easiest): drag FITC + DAPI together.
- **Open FITC... (⌘O)** and **Open DAPI... (⌘D)**.
- If you use the wrong button (e.g. a blue file with "Open FITC"), the program asks: *"... looks like a DAPI image, open it as DAPI?"*
- **Opening a new FITC removes the previous field's DAPI** (another field's nuclei would separate cells wrongly). Open the new field's DAPI too.
- **Close images (⌘W):** clears the window; collected results are kept.

### 6.2 Checking the calibration

The **Calibration** line in the right panel also tells the source: - *read from label* — the value under the scale bar (20 µm, 50 µm) was read. ✅ - *entered manually* — the value you typed into the **Scale bar** box. - *label NOT READ, assumed* — a red warning appears; type the value shown in the image into the **Scale bar** box.

> ⚠️ A wrong scale value makes **area and IntDen wrong several-fold** (e.g. 20 µm instead of 50 µm: 6.25×). Glance at the calibration line for every new image.

Known values for DP74 + cellSens: **20 µm = 138 px → 0.1449 µm/px (40×)**; **50 µm = 172 px → 0.2907 µm/px (20×)**.

### 6.3 Cell-separation indicator

| Indicator | Meaning | What to do |
|---|---|---|
| 🟢 **DAPI nuclei** | Touching cells were separated using nuclei | — |
| 🟡 **Shape only (no touching cells)** | No DAPI, but none needed | — |
| 🔴 **Shape only — N touching cell pairs** | Touching cells separated by shape only (~5% error) | Open the DAPI of the same field |
| 🔴 **DAPI does not seem to belong to the same field** | Most nuclei lie outside the cells | Open the correct DAPI (consecutive numbers can still be different fields) |

### 6.4 Selecting cells

- **Click** a cell → it turns yellow and is measured; click again → deselected.
- **Select all cells** / **Clear selection**.
- Do not measure cells **cut off** at the image border (their area is incomplete).

### 6.5 Background

- **Automatic** (default): **3 boxes** (50×50 pixels) ≥ 40 pixels away from cells, free of signal/debris, as close to the cells as possible. Size and number are set in the right panel.
- **Moving boxes:** hover over a box (hand cursor), press and drag. It is re-measured when released. Moving an automatic box switches to **Manual**; the other boxes stay in place.
- **Manual:** click an empty area → a box is added; click a box → it is removed. **Clear manual boxes**.
- A box overlapping a cell is not accepted (it moves back).
- Per-box values: **Background boxes** tab below.

> ℹ️ **If the background is 0**, the camera/cellSens clipped the dark background to zero; CTCF = IntDen. Comparisons are valid if all images were acquired with the same settings.

### 6.6 Adding to the results table

- **Group / condition:** e.g. *Control*, *Drug 10 µM 24 h*. Plots and statistics group by this label.
- **Experiment / replicate:** independent experiment (biological replicate): *1*, *2*, *3*… Needed for statistics on experiment means (Section 10).
- ➕ **Add to results table (⌘↩).** Re-adding the same image replaces the earlier entry (it is not counted twice).

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## 7. Editing cell boundaries

Automatic boundaries are good in most images; dense clusters, debris or dim cells may need corrections. **Toolbar 2 ▸ Boundary editing**:

| Tool | How | When |
|---|---|---|
| **Select** | Normal mode | Select cells, move boxes, pan |
| **Split** | Draw a line across touching cells **from one edge to the other** | Two (or more) cells were counted as one |
| **Merge** | Click one cell, then its neighbour | One cell was split in two |
| **Delete** | Click the region | Debris or a non-cell region |
| **Draw** | Draw the cell boundary with the mouse; it closes when released | A missed cell or a poorly outlined boundary (the drawing overrides cells beneath) |
| **Undo / Redo** | ⌘Z / ⇧⌘Z (Windows: Ctrl+Z / Ctrl+Y), 30 steps | Every edit can be undone |

- In Split and Draw modes, pan by holding **⌥ Option (Windows: Alt)** while dragging.
- Splitting and merging **preserve the total area**.
- Edits are written to the **Manual_edits** column in Excel (e.g. *merge 1+2; split 1*).
- Changing the scale value re-analyses the image and **resets the edits** (a warning appears below). Calibrate first, then edit.

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## 8. Batch-measuring a folder

**Results ▸ Batch-measure folder... (⌘B)**

### 8.1 Recommended folder layout

```
Experiments/
  Exp_1/
    Control/      ← FITC + DAPI images
    Treatment1/
    Treatment2/
  Exp_2/
    Control/
    ...
```

- **Group** = name of the folder containing the image; **Experiment** = the folder above it.
- If you select a single experiment folder (e.g. `Exp_1`), its name becomes the experiment name.
- Both can be edited in the table before measuring.

### 8.2 FITC–DAPI pairing

Each FITC is paired with the DAPI, among those **within ±3 file numbers in the same folder**, whose **nuclei actually overlap the cells**. Overlap is a **chance-corrected** score (1 = all nuclei inside cells, 0 = no better than random placement); ≥ 0.5 is required. A raw fraction would mislead: in a crowded field where cells cover half the image, a DAPI from another field still overlaps ~50%. Consecutive numbers do not always mean the same field — the program checks the images themselves.

| Status | Meaning |
|---|---|
| 🟢 **ready** | Pair found (percentage in the Overlap column) |
| 🟠 **no DAPI** | No matching DAPI; cells will be separated by shape only |
| ⚪ **skipped: no camera information** | Merged or re-saved image |
| ⚪ **skipped: processed image** | Saved by Photoshop etc. |
| ⚪ **skipped: could not tell FITC from DAPI** | Ambiguous colour (e.g. merged) |
| ⚪ **skipped: unpaired DAPI** | A DAPI that belongs to no FITC |

### 8.3 Measurement

- Ticked rows are measured; untick those you do not want.
- In each image: **all cells** + **automatic background** (box size/number from the right panel).
- **Cells touching the image border** are excluded by default.
- The **Selection** column in Excel records *manual click* or *batch: all cells, edge excluded*.
- After batch measurement, opening suspicious images one by one is recommended (especially dense clusters).

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# 9. Results, Excel and autosave

## 9.1 Excel sheets

| Sheet | Contents |
|---|---|
| **Cells** | All per-cell measurements with context |
| **Group_summary** | n, mean, SD, SEM, median, min, max per group (CTCF) |
| **Image_means** | Image (field) means — ready to paste into GraphPad |
| **Experiment_means** | Experiment means — the recommended statistical unit |
| **Background** | Per-box background measurements |
| **Methods** | Formulas, methods, citation |
| **Validation**, **Validation_summary** | Manual (Fiji) vs automatic comparison and Bland–Altman summary (if present) |

## 9.2 Key columns in the Cells sheet

| Column | Meaning |
|---|---|
| Area, Mean, StdDev, Min, Max, IntDen, RawIntDen | Same definitions as ImageJ |
| BG_Mean, CTCF, CTCF_per_Area | Background mean, CTCF, CTCF/area (= Mean – BG_Mean) |
| Saturated_px | Saturated pixel count (should be 0) |
| um_per_px, Scale_bar_um, Scale_source, Objective | Calibration and its source (label / user / default) |
| BG_mode, BG_box_px, BG_n | Background method |
| Segmentation, Nucleus_method, DAPI_overlap, Touching_pairs | Cell separation details |
| Selection, Manual_edits | How cells were selected; manual boundary edits |
| Fiji_check | Result of the Fiji verification, if done |
| App_version, Timestamp | Program version, date |

## 9.3 Other outputs

- **CSV** (opens correctly in Excel, including non-ASCII characters).

- **ROI image (PNG):** with outlines and boxes.
- **ImageJ/Fiji comparison package (ZIP):** measured channel (TIFF) + ROIs (RoiSet.zip) + instructions (`HOW_TO_USE.txt`).

### 9.4 Not losing your work

- **Autosave:** after every addition, `Documents/ICC CTCF Olcer/otomatik_kayit.xlsx`. Even after a crash or power cut nothing is lost; on restart the program asks *"Continue where you left off?"*
- **Open previous results (Excel)...:** load a saved Excel file and continue measuring (the same image is not counted twice).
- Quitting with unsaved results asks whether to save.

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## 10. Plots and statistics (comparing groups)

**Results ▸ Plot and statistics... (⌘G)**

| Setting | Options |
|---|---|
| **Control group** | Reference for Dunnett's test and normalisation (a group named *Control/Kontrol/untreated/vehicle*... is chosen automatically) |
| **Variable** | CTCF · CTCF_per_Area · Mean · IntDen · Area |
| **Statistical unit** | Cell · Image mean · **Experiment mean (recommended)** |
| **Multiple comparisons** | **Dunnett** (each group vs control) · Tukey HSD (all pairs) |
| **Plot type** | Bar (mean ± SEM) · Box · Violin — all with individual points |
| **Group order** | Drag to reorder (control always first, grey) |
| Normalise | Control mean = 1 |
| Asterisks | GraphPad notation: ns, * p<0.05, ** p<0.01, *** p<0.001, **** p<0.0001 |

**Tests:** 2 groups → Welch's t-test + Mann–Whitney U. 3+ groups → one-way ANOVA + Kruskal–Wallis + Dunnett (or Tukey).

### 10.1 Important: the statistical unit (pseudoreplication)

Cells from the same experiment are not independent. Counting every cell as a replicate makes **p values look smaller than they are** (one of the most common issues flagged by reviewers). Example (simulated data, 3 experiments × 3 images × 6 cells):

| Unit | n | Dunnett p (treatment vs control) |
|---|---|---|
| Cell | 54 | 0.069 |
| Experiment mean | 3 | **0.62** |

**Recommendation:** in publications, run statistics on **means of independent experiments** (at least 3); individual cells can still be shown as points in the plot. The program warns when you test at the cell level.

Dunnett p values are computed by numerical integration; the program uses a fixed seed, so **the same data always give the same p**.

Save: plot as PNG (600 dpi), PDF, SVG; statistics as .txt.

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## 11. Fiji bridge and method validation

**Fiji menu:**

### 11.1 Verify with Fiji (⌘⇧F)

Has **Fiji's own Measure engine** measure the selected cells and background boxes **in the background** (no Fiji window, ~5 seconds) and shows the results side by side with the program's. When the image is added to the results table, the outcome is written to the **Fiji_check** column (e.g. *OK (1.54p99, max diff 3.3e-07 %)*). Changing the selection, boxes or boundaries invalidates the earlier check.

### 11.2 Open in Fiji

Opens Fiji with its interface: image, scale, all ROIs in the ROI Manager, Measure results. Files: `Documents/ICC CTCF Olcer/Fiji/<image name>/`.

### 11.3 Compare with manually traced ROIs (method validation)

To quantify how close the automatic boundaries are to manual tracing (recommended for papers):

1. In Fiji, open the FITC image. Trace a cell with the **Polygon** or **Freehand** tool → **T** (add to ROI Manager). Repeat for all cells.
2. ROI Manager ▸ **More »** ▸ **Save...** → `manual_RoiSet_<image>.zip`
3. In the program, open the same image (and its DAPI) → **Fiji ▸ Compare with ROIs traced manually...** → choose the zip.
4. Each manual ROI is matched to the automatic cell it overlaps most: area difference, CTCF difference, **Dice** and IoU (boundary overlap; 1 = perfect).
5. ➕ **Add to validation table.** Repeat for 3–5 images (separate + touching cells, different conditions).
6. **Fiji ▸ Method validation table and Bland–Altman...:** scatter plot + Bland–Altman (percentage difference), mean difference (bias), 95% limits of agreement, Pearson r, Lin's CCC, Dice.

**Recommendation:** ≥ 30 cells, ≥ 3 images. Optionally re-trace the same cells a few hours later (intra-observer variability): if the automatic–manual difference is not larger than the manual–manual difference, the method is at least as reliable as manual measurement.

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## 12. Warnings: what they mean and what to do

### 12.1 Opening images

| Warning | Meaning | What to do |
|---|---|---|
| *The image appears to have been processed with software (Adobe Photoshop…)* | The file was re-saved by an image editor | Use the raw microscope output |
| *No camera information (EXIF)* | Merged or re-saved image | Use the raw single-channel FITC |
| *… looks like a DAPI image, open it as DAPI?* | File colour does not match the button | Usually **Yes** |
| *The FITC and DAPI images differ in size; DAPI was not used* | The two files are not the same acquisition | Open the correct DAPI |
| *New FITC opened; open the DAPI of the same field* | The previous field's DAPI was removed | Open the new field's DAPI |

### 12.2 Calibration and cell separation

| Warning | Meaning | What to do |
|---|---|---|
| 🔴 *The scale bar value could not be read; 20 µm was assumed* | Label not read | Type the value from the image into the **Scale bar** box |
| *No scale bar found — pixel units* | No scale bar | Area in pixel²; comparisons valid only at identical magnification |
| 🔴 *Cell separation: shape only — N touching cell pairs* | No DAPI, touching cells present | Open the DAPI of the same field |
| 🔴 *This DAPI image does not seem to belong to the same field* | Most nuclei lie outside the cells | Open the correct DAPI |
| *Re-analysed: previous manual edits were reset* | Scale etc. changed | Redo the edits |

### 12.3 Background

| Warning | Meaning | What to do |
|---|---|---|
| ℹ️ *All pixels in the boxes are 0* | Camera black level clipped; CTCF = IntDen | Fine if all images used the same settings |

| Warning | Meaning | What to do |
|---|---|---|
| ⚠️ *Some boxes may contain signal/debris* | Bright pixels in a box | Drag the box to a clean area |
| 🔴 *No background box — CTCF cannot be computed* | No room for boxes / no box in Manual mode | Reduce the box size or add boxes manually |
| ⚠️ *The box overlaps a cell* | Boxes cannot be placed on cells | Place it a little further away |

### 12.4 Measurement table

| Situation | Meaning | What to do |
|---|---|---|
| **Saturated px > 0** | Pixels at the camera's maximum → CTCF is **underestimated** | Re-acquire with lower exposure, or exclude that cell |

### 12.5 Statistics

| Warning | Meaning | What to do |
|---|---|---|
| ⚠️ *Each cell was counted as a replicate (risk of pseudoreplication)* | n = number of cells | Use the **Experiment mean** unit for publication |
| ⚠️ *Different objectives are mixed (20×, 40×)* | Groups acquired at different magnifications | Compare only images from the same objective |
| ⚠️ *Zero within-group variability* | Identical values repeat (same image added twice?) | Check the table; the test is not reliable |
| ℹ️ *2 experiments per group: low statistical power* | n < 3 | At least 3 independent experiments |
| *Statistics require at least … per group* | Not enough data | Add more images/experiments |
| ⚠️ *Few cells: r and CCC are meaningless* (validation) | Too few cells for Bland–Altman | ≥ 30 cells, ≥ 3 images |

### 12.6 Batch measurement and Fiji

| Warning | Meaning | What to do |
|---|---|---|
| 🟠 *no DAPI (best candidate …%)* | No overlapping DAPI found | Put the correct DAPI in the folder, or accept shape-only separation |
| ⚪ *skipped: …* | See 8.2 | — |
| *Fiji was not found* | Fiji not installed / elsewhere | Fiji ▸ Choose Fiji location… |

| Warning | Meaning | What to do |
| --- | --- | --- |
| ⚠️ *Difference found* (Fiji verification) | Program and Fiji differ (not expected) | Check calibration; please report the issue |

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## 13. For your paper: methods text and citation

**Example methods text (adapt to your settings):**

> Corrected total cell fluorescence (CTCF) was quantified as CTCF = integrated density − (cell area × mean background fluorescence) (McCloy et al., 2014) using ICC CTCF Ölçer v2.0.0 (Balcan, 2026; https://doi.org/10.5281/zenodo.23860021), an open-source tool whose measurements are identical to ImageJ/Fiji for the same regions of interest (verified with Fiji 1.54p). Images were acquired on an Olympus BX43 fluorescence microscope with an Olympus DP74 camera (cellSens Entry); FITC fluorescence was measured on the green channel of the unprocessed images (40× objective, 0.145 µm/pixel; calibration read from the embedded scale bar). Cells were segmented from the FITC channel (Li threshold) and touching cells were separated by watershed seeded with DAPI-stained nuclei. Background was the mean of three 50 × 50 pixel cell-free regions per field. Cells touching the image border were excluded. Automatic boundaries were compared with manual Fiji tracing in n = … cells from … images (Bland–Altman bias …%, 95% limits of agreement … to …%; mean Dice …). Statistics were performed on the means of N = … independent experiments using one-way ANOVA followed by Dunnett's test versus control.

The **Methods** sheet in Excel and the *App_version, Nucleus_method, Scale_source, BG_box_px, BG_n, Selection, Manual_edits, Fiji_check* columns in the **Cells** sheet contain everything needed to fill in this text.

**Citation (reference list):**

> Balcan E. (2026). *ICC CTCF Ölçer: per-cell corrected total cell fluorescence (CTCF) quantification for immunocytochemistry images* (Version 2.0.0) [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.23860021

| DOI | When to use |
|---|---|
| **10.5281/zenodo.23860021** | **This version** (2.0.0) as used in your paper — recommended for reproducibility |
| 10.5281/zenodo.23860020 | **All versions** of the program (concept DOI; always resolves to the newest version) |

The version used is recorded in the **App_version** column in Excel; if you measured with a newer version, take that version's DOI from the Zenodo page (https://doi.org/10.5281/zenodo.23860020 → "Versions").

For the CTCF method: McCloy RA, Rogers S, Caldon CE, Lorca T, Castro A, Burgess A. (2014). Partial inhibition of Cdk1 in G2 phase overrides the SAC and decouples mitotic events. *Cell Cycle* 13(9):1400–1412. https://doi.org/10.4161/cc.28401

## 14. Troubleshooting

| Problem | Solution |
|---|---|
| Mac: *"unidentified developer"* | Right-click the app ▸ Open ▸ Open (once) |
| Mac: *"is damaged and can't be opened"* | Terminal: `xattr -cr "/Applications/ICC CTCF Ölçer.app"` |
| Mac (Apple Silicon, Monterey): does not open | Update macOS to 12.3 or later |
| Windows: *"Windows protected your PC"* | More info ▸ Run anyway |
| Slow first start | A font cache is built on first launch (once, ~10–30 s) |
| Wrong cell count | Did you use the right button (FITC/DAPI)? Is the DAPI from the same field? Correct boundaries as in Section 7 |
| A dense cluster counted as one "cell" | **Split** tool |
| Nuclei (magenta outlines) look wrong | Switch to **Display: DAPI** and check; is the DAPI from the same field? |
| Values differ slightly from v1 | v2 uses different nucleus detection (Nucleus_method). Use one version throughout a study |
| Excel cannot be saved | If the file is open in Excel, close it and retry |
| The program closed / crashed | Reopen → *"Continue where you left off?"* → Yes (autosave) |
| Fiji bridge does not work | Fiji ▸ Choose Fiji location… ; open and close Fiji once yourself (first-run updates) |
| Interface in the wrong language | View ▸ Language / Dil, then restart |

Questions and bug reports: https://github.com/erdal-b-dot/icc-ctcf-olcer/issues

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## 15. Technical details

**Measurement definitions (same as ImageJ):** - Area = number of pixels × (µm/pixel)² - IntDen = Area × Mean (calibrated units); RawIntDen = sum of pixel values - StdDev = sample standard deviation (n – 1) - ROI rule: a pixel belongs to the ROI if its centre lies inside the polygon - CTCF = IntDen – Area × BG_Mean; CTCF_per_Area = Mean – BG_Mean

**Calibration:** the length of the cellSens scale bar (white box, lower right) is measured in pixels; the bar is located first (a thin black line with white above and below), the box is measured from the bar's white margin, and the number below it is read with digit templates. Tested on 3,534 fluorescence images (author's archive 3,233 + development set 301; two cellSens box styles): all read correctly. Objective = camera pixel size (DP74: 5.86 µm) ÷ µm/pixel.

**Cell segmentation:** FITC (G channel) Gaussian smoothing → Li threshold → closing, hole filling, small-object removal (< 50 µm²) → watershed seeded with DAPI nuclei (distance transform if no DAPI).

**Nucleus detection (Nucleus_method):** | Method | Version | Description | |---|---|---| | `otsu` | v1.0.0 | Otsu threshold (also includes the dim DAPI halo) | | `bright_core` | early v2 | Upper threshold of 3-class Otsu (misses dim nuclei) | | **`local`** | **v2.0.0 default** | Otsu mask → watershed from smoothed DAPI maxima → each nucleus trimmed at half of its own peak intensity |

**Validation results:** - Identical to Fiji 1.54p for the same ROIs: Area/Mean/StdDev/Min/Max/IntDen/RawIntDen (5 ROIs in a 40× image; 54 ROIs in a 20× image with non-zero background; largest difference < 0.00001 %). - Manual tracing (1 image, 2 touching cells — preliminary): Dice 0.943 / 0.938; CTCF difference +2.3 % / +3.7 %. - Dense clusters at 20×: 47 nuclei, 51 cells detected.

**Program files:** `core.py` (measurement), `batch.py` (batch), `report.py` (Excel, plots, statistics), `fiji.py` (Fiji bridge), `desktop.py` (interface), `i18n.py` + `locales/` (languages), `tests/test_core.py` (validation tests).